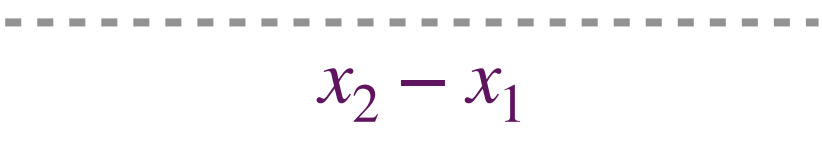


© Zahra Shakeri, Dalla Lana School of Public Health, University of Toronto



Institute of Health Policy, Management and Evaluation
UNIVERSITY OF TORONTO

Dalla Lana
School of Public Health



$$x_2 - x_1$$


$$y_2 - y_1$$

1

5

DISTANCE IN KNN (EUCLIDEAN DISTANCE)

For two points in a plane with coordinates (x_1, y_1) and (x_2, y_2) , the Euclidean distance between them is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(x_1, y_1) \odot$$





$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$





$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

, the distance is:

and

Example-Fortwo points in 3D space

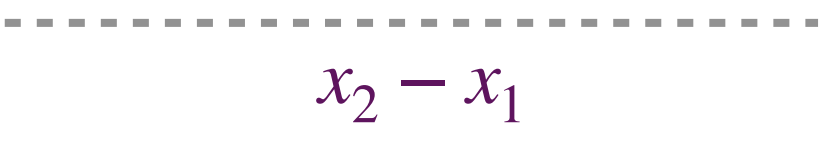
(x_1, y_1, z_1)

(x_2, y_2, z_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



MODREATAURRESINOURDATA



$$x_2 - x_1$$


$$y_2 - y_1$$



$$(x_1, y_1) \odot$$





$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$